Semiconductor Device Aging

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Abstract:

Semiconductor devices are some of the most critical technological devices in modern society, used to power and operate a large array of electronics that are deemed as essentials to the everyday lives of hundreds of millions. Given the importance of these devices, their optimization has been an integral area of study, which predominantly discusses the aging process of these devices. Therefore, this paper delves into the issue of semiconductor device aging, beginning with an introduction to semiconductor devices and what they exactly are, along with what it means for semiconductor devices to age. Causes of aging will then be discussed, along with physical and societal impacts of these aging processes with reinforcing case-studies. This paper will then discuss how these aging processes are measured and what tests are conducted in order to regulate the integrity and reliability of these devices. Finally, semiconductor device aging mitigation strategies will be covered, and conclusions are made regarding the significance of this issue and its research/advancements.

Introduction:

Semiconductors are a component of electronic devices which are made from pure "semiconductor" elements, such as silicon, germanium, or gallium arsenide as common examples. These elements are considered semiconductor materials because when they are methodically exposed to small amounts of impurities, known as doping, the conductivity of these elements changes drastically, allowing them to behave both as a conductor and as an insulator when desired. This makes semiconductor devices so important in today's technological world, as these devices/components essentially comprise all of the electronic devices in every major industry that exists today, such as healthcare, communications, military systems, transportation, computing, etc., generating hundreds of billions of dollars globally each year. Imagine what our world would look like today without our everyday devices such as smartphones, computers, televisions and without crucial technologies such as advanced medical equipment? These devices are the brains of our current technology, and therefore their unfavorable characteristics, such as their aging processes, are a focal point of modern day research. In today's world, semiconductor devices are so integrated into daily life that they theoretically are an extension of ourselves.

Semiconductor device aging refers to the loss of electrical characteristics of a semiconductor device over time, leading to gradual degradation of the device's performance and reliability. As technologies become more advanced, the levels of integration and conditions that semiconductor devices have to operate in have become more demanding, leading to greater stress and pronounced degradation of these devices electrical properties, flattening the curve on technological progress. Therefore, being able to understand the aging process of semiconductor devices is paramount to the efficiency and overall performance of all modern day electrical devices. The deeper our understanding of the aging process becomes, the greater the potential to improve the performance, reliability, and longevity of emerging electronic devices, thus dismantling obstacles to technological progress.

Semiconductor Device Aging:

Semiconductor device aging is a major concern as society becomes more reliant and absolutely dependent upon electrical devices. There is not a singular way in which semiconductor devices age, as there are a multitude of physical and chemical processes within a device that contribute to the natural phenomena of the degradation of a device's electrical characteristics. For this reason, aging becomes a complicated issue as there are many different sources of potential aging processes within a semiconductor device, and given their interconnectedness, provides exponentially more problems for engineers to try and solve.

One major mechanism in which semiconductor devices age is through Bias Temperature Instability (BTI). BTI includes two different types of instability, Negative-Bias Temperature Instability (NBTI) and Positive-Bias Temperature Instability (PBTI). NBTI is a typical mechanism found in PMOS transistors, and similarly, PBTI is typically found in NMOS transistors. Typically, a PMOS transistor will experience different types of NBTI depending on what state the transistor is operating in. One of these states is "the 'stress' phase, which occurs when the transistor is on, [where] positive interface traps are generated at the Si-SiO2 interface" [1]. This process can lead to an increase in the threshold voltage, Vt, of the transistor over time, as when these positive interface traps are made, they act as electron traps which essentially "capture" and immobilize electrons at the gate oxide region. Over time, these positive traps will accumulate and create localized positive charges at the interface, making the gate oxide "thicker" and therefore requiring a greater threshold voltage in order to reach the desired channel conductivity and have the transistor perform its function. The other phase, which is known as "'recovery', occurs when the transistor is off. In this phase, the threshold-voltage drift occurred during the stress phase will partially recover" [1]. Although the increased temperatures of the stress state are no longer present, due to the damages of the stress phase, the MOSFET may recover some of its original threshold voltage level but will never return to its original level due to BTI. This applies both for PMOS and NBTI and NMOS and PBTI, respectively, and PBTI is very similar to the processes of NBTI. Rather than a negative bias voltage being applied at the transistor gate terminal, it is a positive bias voltage due to the nature of the NMOS gate dielectric and semiconductor substrate. This would lead to negative charges building up at the gate oxide and interface as opposed to the positive charges of NBTI. Although NMOS is much more commonly used than PMOS transistors, "early attention was focused on NBTI, as it is generally higher than PBTI" [2]. But, although PBTI was not as significant at the time due to interface states not being created, as the development of high-k/SiON stacks became more prevalent in the semiconductor industry, so did the significance of PBTI. As can be seen in Figure 1, when modern commercial semiconductor processes with high-k/SiON stacks were measured during stress and recovery, it is apparent that the recovery for PBTI was much less than that of NBTI, resulting in almost twice as much change in the threshold voltage over the time measured "post recovery". This illustrates the significance of this aging process, as the threshold voltage is a significant characteristic of properly performing transistors, as if this characteristic is heavily manipulated under temperature changes (which are inevitable when operating electronic

devices), this aging process also becomes inevitable. Although it may be less significant than in the early development of MOSFETS, bias temperature instability still impacts our modern day transistors performance and reliability.

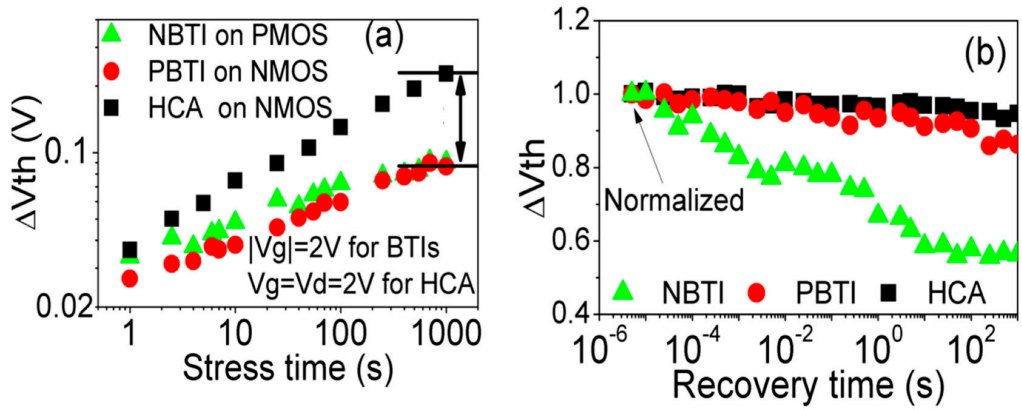

Figure 1: A comparison of PBTI with NBTI during stress (**a**) and recovery (**b**). (**a**) Shows that the PBTI was similar to NBTI during stress for this CMOS process but more stable during recovery. [2]

Another related mechanism in semiconductor device aging is the process of Hot Carrier Injection (HCI). Similar to that of BTI, HCI also impacts the threshold voltage, but does so in a different process. First, it begins with hot carrier generation. The question then arises, how exactly are these hot carriers "generated"? This process begins when the gate voltage, V_G, is comparable or lower to the voltage from the drain to the source, V_{DS}, and the drain voltage is greater than the source voltage (V_D>V_S). When this happens, the voltage drop due to the channel current is concentrated on the drain side, as can be visualized in Figure 2 and in the Figure 3 boxed visual. This leads to these carriers that are traveling from the source to the drain to gain a huge amount of energy as they go through the high-field region, resulting in their kinetic energy becoming much higher than the lattice temperature and becoming "hot".

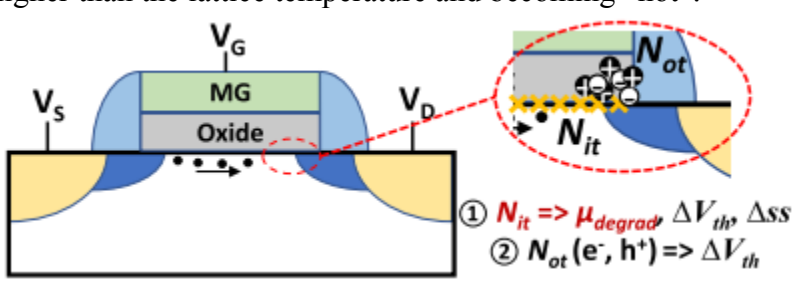

Figure 2: Schematic representation of the hot-carrier-induced damage in the MOSFET, which typically localizes near the drain side due to the carrier energy peak there. Inset shows the hot-carrier-induced interface state Nit and oxide charge trapping Not: for the latter one, both electron and hole (from impact ionization) trapping can occur, depending on the carrier energy and local oxide field. [4]

Now, why does this lead to hot-carrier aging? This is due to the fact that "some of the hot-carriers gain so much energy (about 3.2eV for electrons and 4.7eV for holes) that they can surmount the energy barrier at the Si/SiO2 interface and be injected into the oxide, producing a small gate current (IG)" [3]. When this happens, sometimes a small portion of the injected hot

carriers get trapped in the oxide interface, and if these carriers have enough energy, they can even break bonds at the oxide and SiO2 interface. This is what leads to breakdown in these interfaces, and with enough injections into them over long periods of time, the damages become permanent changes, altering the MOSFET and IC's operation. As can be seen in the tree diagram in Figure 3, these cascading effects result in a multitude of changes in the MOSFET, from the threshold voltage to the drain current to the subthreshold slope. This illustrates yet another major aspect of device aging, as not only can bias temperature instability of the transistor in operation result in damage over time, but so can the hot carriers that are created by the source-gate-drain interface. This leads to engineers not only having to worry about the bias temperature of the MOSFET but also have to be aware of the values of the gate, drain, and source voltages at all times to minimize the occurrence of hot carrier generation.

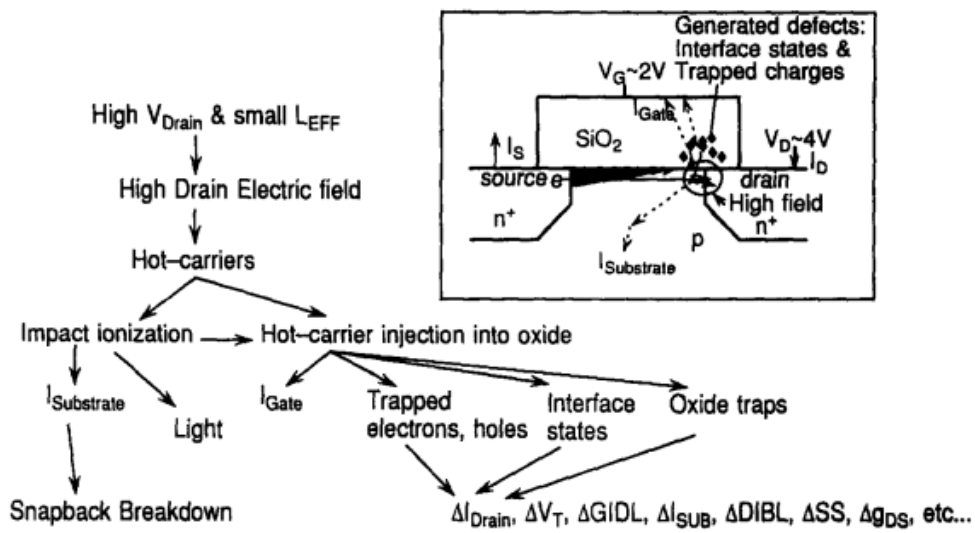

Figure 3: Summary of the hot-carrier generation and degradation in MOSFETs. The inset shows a NMOSFET under the typical hot-carrier stress conditions and the resulting high-field region of the drain. [3]

Although the previously mentioned issues are very significant, both contribute to damage to specific electrical characteristics of the transistors, such as their threshold voltage. Time-Dependent Dielectric Breakdown (TDDB), on the other hand, is a mechanism of aging that results in stress breakdown of the MOSFET and its oxide layer as a whole. When a MOSFET is under continuous electric field stress, this sustained stress results in a multitude of breakdowns, which in their entirety lead to the dielectric material's full breakdown as a whole over a given time period, hence the name. One process that occurs during sustained electric field stress is the trapping of electrons and holes in the gate oxide, similar to BTI and HCI. When these electrons and holes become trapped, a multitude of breakdowns and characteristic changes can arise. Under high oxide field conditions, hole trapping dominates over electron trapping, and when this occurs, as seen in Figure 4, the threshold voltage reduces and the leakage current at the gate of the MOSFET rises until eventual breakdown. This illustrates how hole trapping leads to an increase in F-N tunneling currents, which causes another form of breakdown in electron

tunneling. When these electrons undergo quantum mechanical tunneling through potential energy barriers, this leads to unwanted creation of electron hole pairs given the presence of electrons that were not meant to be present within the dielectric. These pairs then create new conducting channels within the dielectric that are undesired, leading to channel paths that disrupt the MOSFET channel and breakdown its function. On the other hand, under lesser oxide field conditions, as seen in Figure 4, electron trapping dominates, and although this also leads to an increase in threshold voltage, it lowers the tunneling and leakage current, lowering the overall stress and damage on the MOSFET system. This illustrates how under different oxide electric field conditions, there are different mechanisms to MOSFET failure, and therefore dielectric breakdown is either accelerated or minimized. As can be seen from Figure 4, "oxide degrades faster under higher oxide fields" [5]. For that reason, engineers tend to work in lower fields when measuring TDDB in order to get a more accurate measurement of the lifetime of the device under normal operating conditions, as "lifetime predictions can be significantly overestimated[…] if the TDDB measurements are conducted only at high oxide electric fields" [5]. All in all, with an increased electric field stress, threshold voltage change, leakage current, electron tunneling, all combine to create conducting paths between electrodes, breakdown paths, short circuits, and ultimately catastrophic failure of the MOSFET device.

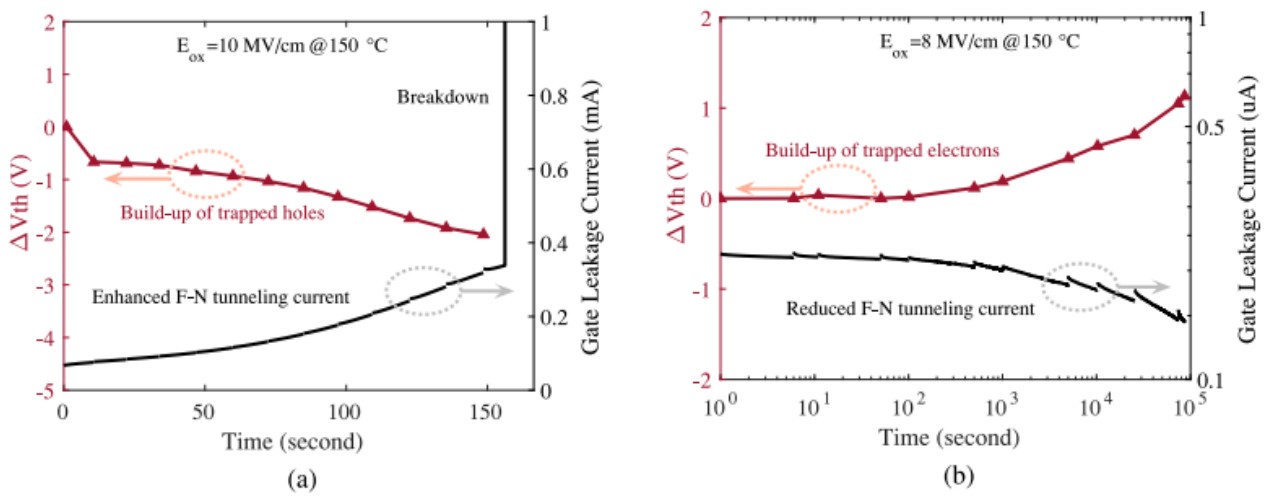

Figure 4: . Gate leakage current behaviors and threshold variations of vendor E DUTs under constant gate voltage stress with oxide electric fields of (a) 10 MV/cm and (b) 8 MV/cm. The trends of the threshold voltage shifts reflect the build-up of the trapped charges, while the behaviors of the gate leakage currents indicate the effects of the trapped charges on the F-N tunneling currents. [5]

One final mechanism of semiconductor device aging that is very significant and highly researched is the process of electromigration. In this phenomena, as the name suggests, is the literal "migration", or transport of matter in a MOSFET from one place to another. Specifically, in MOSFETs, metal interconnects are used to connect different transistors, capacitors, and other components on the integrated circuit. When high dc current density flows through these metal interconnects, typically when a device is in its "on" state, these moving electrons can collide with the metal atoms, causing them to move with the flow of electrons. When these atoms move,

they begin to create a void in their previous location as it is now absent. The build-up of these voids lead to literal physical holes and eventual cracks in the interconnect structure, as can be seen in Figure 5. These cracks are a critical issue, as they can disrupt the flow of signals and form open circuits within the device, increasing the overall resistance among the electrical paths and therefore higher voltage drops.

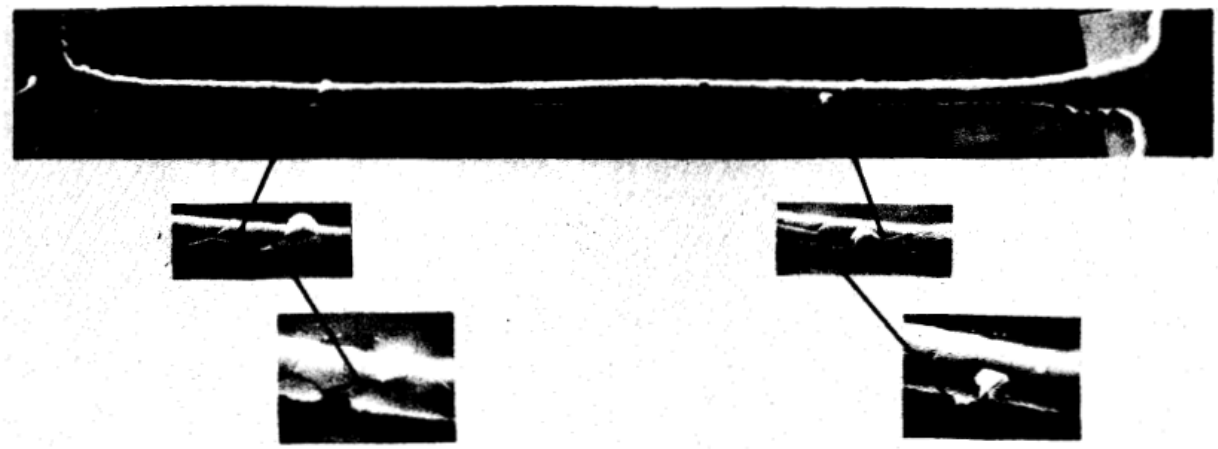

Figure 5: Defects due to electromigration in a large grain aluminum stripe tested for 210 h at 170°C and of 2 x 10^6 A/cm^2 [6]

It is just as important to acknowledge, however, the fact that this migrating matter not only creates holes where it once was, but also "causes accumulation of matter in the form of hillocks and whiskers which can have very detrimental effects" [6]. These effects include, but are not limited to "shorts between superimposed conductive stripes separated by a dielectric layer, or between adjacent stripes[...] [and] accumulation of aluminum at a silicon contact [which] could cause fracture of a protective layer which would then lead to failure by corrosion" [6]. This illustrates that holes and cracks are not the only critical failures of electromigration, but so is the accumulation of the migrated matter in another location on the substrate. This highlights the multitude of critical challenges that may emerge when utilizing semiconductor devices over extended durations and amid intense operational stresses.

Impacts of Semiconductor Device Aging:

Being aware of the many forms of device aging, the question now arises, what do these failures truly mean in a physical sense for the semiconductor device? As partially discussed in the previous section, these aging mechanisms and processes are created due to the physical disturbances they create within the system. One aspect that was highly covered already is that of the threshold voltage, which is the minimum voltage required to create a conducting channel between the source and drain within a MOSFET, allowing its operation. When the threshold voltage of a MOSFET is impacted, it can no longer be determined what amount of power will activate/deactivate the MOSFETS within the device, therefore losing control of the device's operation and therefore reliability. For example, if it takes more power to operate a battery powered device due to aging raising its threshold voltage, then its battery life will become

shorter and shorter, something commonly experienced by individuals with battery powered devices over time. Similarly, aging mechanisms that result in increased V_{th} typically are inversely related to another physical impact on MOSFETS, leakage current. Leakage current is the current that flows through a transistor when it is in its "off" state, usually in the forms of subthreshold leakage and gate leakage. Defects in an oxide layer and tunneling are the primary candidates to gate leakage, and a decreased V_{th} contributes to greater subthreshold leakage. The more leakage current that occurs in a MOSFET device, the more unwanted power dissipation occurs, and therefore the hotter the temperature of the device becomes. This leads to even more stress and damage, less signal integrity, and an overall lower performance device. These physical changes also directly contribute to another physical change of carrier mobility, as with changing V_{th} and leakage current, the ability for carriers to move around the semiconductor material also changes. The more power being consumed and unnecessarily dissipated by the device, the more stress the device will be under, and the harder it will be for carriers to properly move through the conducting channel making device operation less efficient. With all of these physical changes in a semiconductor device, the increased stress and damage to dielectric material will also lead to lowering the breakdown voltage of the device, which is the voltage in which the device will experience a huge spike of current. This will lead to semiconductor breakdowns such as punch-through, oxide breakdown, avalanche breakdown, and zener breakdown; all which can lead to permanent device damage and failure.

These failures are so important not because of their physical consequences, but rather their socioeconomic impacts. In the end, semiconductor devices are only important due to the value that they are able to provide to human lives, and consequently the amount of money they generate in the economy. As a matter of fact, there are a multitude of modern events in which the failure of semiconductor devices due to rapid aging caused critical socioeconomic consequences. One such event occurred with Microsoft around 2005-2009 regarding their new gaming console product; the Xbox 360. The Xbox 360, which was released in 2005 by Microsoft, was an advanced gaming system which was highly anticipated, selling millions of consoles within its first year of release. But, not too long after their release, gamers were not satisfied and many were wishing to be refunded after their devices no longer worked less than 2 years after their purchase. As can be seen in Figure 8, 23% of Xbox 360's purchased experienced failure within 2 years of purchase, heavily eclipsing their competitors. But, it could also be recognized that without RROD, Xbox 360's failure rate would be quite close to that of their competitor Sony's PS3. Now what was RROD? This was a systematic issue known as the "Red Ring of Death", where 360 consoles would output a red ring on their power button, rendering the console unusable. This was due to the GPU and CPU within these consoles overheating, and therefore experiencing rapid aging effects, leading to total console failure in a short period of time. Due to this issue, Microsoft "announced a $1.15 billion fund to be set aside for Xbox 360 repair and replacement" [7]. Not only was a major company short of billions of dollars, but millions of consumers were also impacted negatively in their daily activities, all due to the consequences of semiconductor device aging.

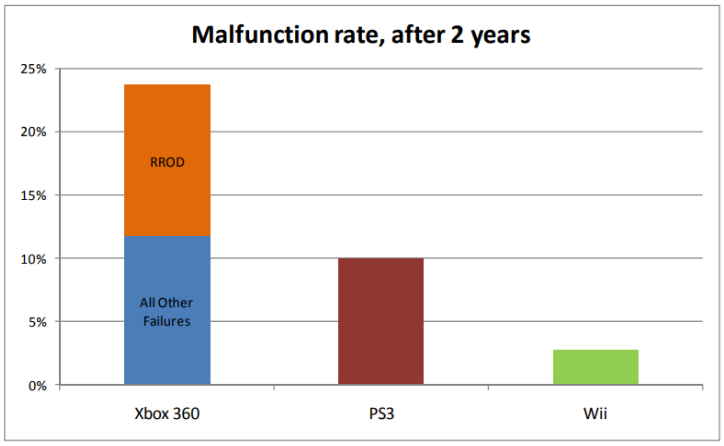

Figure 6: Console failure rates after 2 years of ownership. [7]

These aging processes can also have much more dire consequences. This can be seen with Toyota in 2009-2010 and their recall of vehicles with unintended acceleration. "On August 28, 2009, Mark Saylor, an off duty California Highway Patrol officer driving on Route 125 in a 2009 Lexus ES350 sedan on loan from his Toyota dealer, lost control of his car as it accelerated to speeds of 120 mph. Saylor, his wife Cleofe, their 13-year-old daughter Mahala and 38-year-old brother-in-law Chris Lastrella were killed instantly" [8]. This was a horrible occurrence only highlighting the importance of rigorous device testing and regulation, as this would not be the first or last time the automotive industry would experience recalls due to failures such as this one. Although there were many other mechanical contributing factors to the unintended acceleration, the primary issues fell under the aging and degradation of the semiconductors within the ECUs, leading to an uncontrollable throttle within the vehicles. Due to these horrific events, "between January 19th and February 23rd [2010], Toyota lost $35 billion in stock value" [8], further illustrating the massive impact of semiconductor devices on socioeconomics. When semiconductor devices fail, people lose lives and economies lose billions of dollars.

How to Measure and Test Semiconductor Device Aging:

Given the sheer importance of the reliability of these devices, it is just as critical that there are effective testing and measurement protocols in place to ensure the characteristics of these devices before they are used commercially. One major and effective form of testing that is used to measure the durability and reliability of these devices are accelerated aging tests. In these many forms of accelerated testing, long-term use and environmental stresses are simulated over a short period of time, and the strategy of this is done by "the application of electrical overstress at fixed junction temperature (T_j) in order to avoid thermal cycles which generate package related failures" [9]. In this way, the electrical stress on the system is maximized while minimizing the influence of thermal fatigue on the measurements by preventing thermal cycling. With this, engineers are able to effectively measure threshold voltage, leakage current, drain current, breakdown voltage, time-to-failure, etc. under these accelerated conditions. There are multiple

parts of the accelerated aging system, and that is its electrical operation unit, its in-situ measurement unit, and its thermal block. While the operation unit electrically runs the machine with power supplies and function generators and the measurement unit measures the parameters and data, the most critical section of this device is its thermal block. As mentioned before, the most critical part of the accelerated aging test is "apply[ing] electrical overstress to devices without inducing thermal cycles that generate package related failure mechanisms" [9], and this is what the thermal block achieves. As seen in Figure 7, the thermal block uses a copper block for a large thermal capacitance and a heat sink to direct the heat flow of the system, as well as a thermo-electric unit to electrically control the heat flow. In this way, thermal cycles are able to be prevented, and semiconductor devices can be measured of their aging parameters over hours to days as opposed to years.

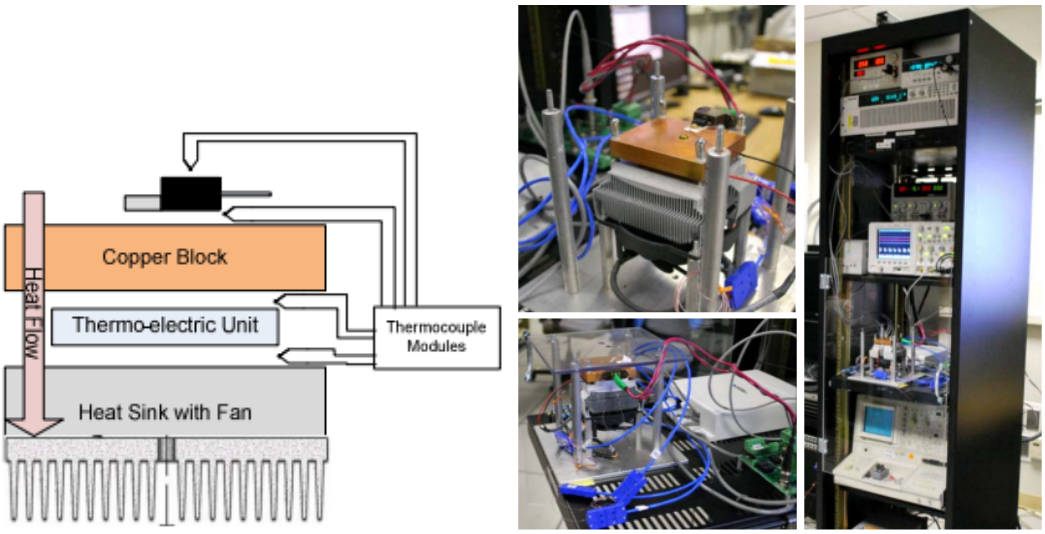

Figure 7: Thermal block for measurement and control of device temperature (left). Images of real thermal block (middle). Image of real full accelerated aging system (right). [9]

An additional measurement process that was touched on previously and is important to mention is that of TDDB measurements. These measurements are so critical for semiconductor devices because they show the reliability and integrity of the dielectric materials, such as gate oxides, within these devices. This is so important as these materials are what are most able to effectively measure the anticipated lifetime of a semiconductor device, so in some way, it could be said that TDDB measurements illustrate the characteristics of a given device more than anything else. By applying an electric field to a device and monitoring the device's breakdown, engineers are able to simulate the device under what are effectively operating conditions for the device. But, as mentioned briefly previously, " TDDB measurements need to be done at oxide electric fields lower than 8.5 MV/cm to avoid hole trapping and ensure more reliable lifetime predictions" [5], as lifetime predictions will be otherwise overestimated. The issue with this, though, is that measuring in lower oxide electric fields equivalently leads to a much longer testing time, which becomes infeasible. In order to combat this, "higher temperatures can be applied to shorten the measurement time at the lower oxide electric fields without compromising

the lifetime prediction accuracy" [5], as the fields where the data is abruptly impacted are independent of temperature changes for a wide celsius interval. In this way, engineers are able to effectively measure the lifetime predictions of devices under a commercially reasonable amount of time.

All in all, these forms of testing in measurement all fall under the concept of reliability modeling and simulation, which is the general rule of thumb that is used for measurement of semiconductor devices and their aging processes. Mainly, these modeling and simulation tests are then used for "three classes of aging detection and prevention schemes [...] workload monitoring, circuit monitoring and periodic testing" [1]. While workload monitoring proactively detects the patterns, tasks, and loads of a given device, circuit monitoring reactively detects results of electrical signals, operation parameters, and reliability performance metrics after the device is put through some form of testing. This leads into periodic testing, as circuit monitoring is not effective without the concurrent testing and collection of data with software and machines such as the accelerated aging chamber. Using software to then model and display this data is critical for engineers to better understand what they are looking at, and make decisions/changes regarding the devices they are working with before they are used commercially. Ultimately, the testing stage of these devices is critical, as it is important that the reliability and integrity of these devices remain intact before they are used in real-life devices, where device functionality could be the difference between life and death.

Preventative Actions to Slow/Prevent Semiconductor Device Aging:

As engineers, the goal is "kaizen", also known as continuously improving every day. Given these aging issues exist and are quite paramount to the semiconductor industry, learning how to minimize and prevent them is something engineers have and will continue to focus on as long as semiconductors are used for our electronic devices. For all of the mechanisms of aging discussed thus far, there are integral techniques that are used to mitigate each issue. Beginning with bias temperature instability, there are several mitigation techniques used, one major one being node activity optimizations. In this technique, engineers focus on reducing the workload on specific nodes and components within a device, nodes which testing had revealed contribute largely to BTI degradation. By doing so, it has been revealed that "a 2-4× reduction in performance degradation, caused by NBTI, can be achieved" [11]. Another technique used specifically for BTI is configuration bit inversion. In this technique, bits within the semiconductor device are periodically flipped in their state, which also periodically changes the areas of high stress within the individual resistors. By doing so, the stress on areas where BTI degradation is building up is reduced, consequently reducing the aging effects on the device as a whole, and it was studied that " a decrease of 2.5% on Failure-in-Time (FIT) [was] observed" [11]. Similarly, operations such as being able to effectively run the devices at lower temperatures and remapping the device's components based on stress results, could allow for the device to experience less overall stress from heat and its oxide electric field, allowing it to withstand for longer periods of time. Remapping of components is also the main mitigation technique for hot

carrier injection, as redistribution of the workloads can directly reduce localized heating effects and cause less hot carriers to be made. In fact, remapping for HCI in studies "result[ed] in around 28% increase in lifetime but with about 1.53% performance degradation" [11]. Regarding time dependent dielectric breakdown, current leakage optimization is an effective mitigation technique. By engineering the properties of the gate dielectric and its thickness, V_{th}, etc., the breakdown of the dielectric will be reduced, and therefore the time to failure will improve. According to research by engineers performing current leakage optimization and measuring TDDB, they "report[ed] 24% improved MTTF [mean time to failure] on average for different benchmark circuits" [11]. Finally, for electromigration mitigation, rerouting is a strong mitigation strategy similar to remapping, where interconnects between resources on the device are literally physically rerouted. In doing so, the routing paths can be continuously more optimized and the current densities can be redistributed, lessening the electromigration effects on a device. In continuously exercising all of these techniques on modern day semiconductor devices, the aging effects on semiconductors will only become more and more improved, as the materials and designs of these semiconductors continue to relieve system stress.

Generally, it can be seen that these techniques are effective, and this is due to the fact that aging adaptation and mitigation techniques fall under four main categories. The largest and one of the most critical concepts is that of worst-case design. The idea of worst-case design is to design a semiconductor keeping in mind what the possible worst conditions of every aspect of the device can possibly be when designing it, even if this worst case scenario has a 0.01% chance of happening. Worst-case designing in semiconductors can be seen through redundancies in case of component failures, as well as guardbanding and designing components such as gate size to withstand minimum requirements and maximum stress conditions. Worst-case designing is arguably the most important design approach of all mitigation techniques, since if the worst case scenarios are able to be effectively mitigated, then device failures will theoretically become obsolete, regardless of aging events. Although not as generally practiced as worst-case design, the approach of design time aging-aware balancing is also one of great importance specifically to the semiconductor industry. When devices are created, their components may be balanced at first in terms of aging effects. But, as the device is used, certain transistor components may experience higher stress levels than others, leading to faster degradation and failure, when all other device components are working smoothly. For this reason, design-time aging-aware balancing addresses this "by balancing the delays of system components at their expected lifetime rather than design-time according to their aging rate" [10]. With these delays, engineers are able to ensure that these devices are able to perform optimally throughout their operational lifetime, as opposed to being alive but dysfunctional due to failed components. This segways into the next category of mitigation approach known as dynamic adaptation. Dynamic adaptation is essentially similar to that of a closed feedback loop, where the device continuously gives feedback to itself in response to its changing conditions and environments, allowing it to change its behaviors or parameters in response. For example, being able to dynamically change the voltage, frequency, or cooling power during operation is critical in minimizing the aging damage

done on the device. If the device were becoming too hot and building up too much thermal energy, the dynamic response of these conditions to the device would cause it to increase the system's cooling power, consequently reducing the thermal energy and the overall system aging damage. Very similarly, the final approach of adaptive resource management also works on the device dynamically, but instead works on the resources of the device rather than the characteristics. For example, adaptive resource management could be doing this such as saving device power when it is in an idle-time, or selectively shutting out power to certain components within the device when they are not in use in order to save power for the components in operation. While this can improve aging effects by allowing components to rest, it also just as importantly makes these devices more energy efficient, as all resources are dynamically allocated.

With all of these mitigation techniques and preventative measures, it must also be remembered that the semiconductor industry is an economy and a business. At the end of the day, businessmen and engineers must give-and- take where necessary in order to achieve their design and monetary goals. For example, much more rigorous testing will be done on the semiconductor devices going into a Boeing airplane than that of those going into a children's toy. While it is important to be aware of and understand trade-offs, it must also be understood that businesses and engineers should not cut corners and pass-up on requirements due to high costs. Unironically, during the period of years after the 2008 recession for example, there were a lot of technological failures in the aviation and automotive industries, most likely stemming from the economic burdens of the recessions, leading to less rigorous testing and a severe lack of the mitigation techniques that were mentioned previously, such as worst-case designing and concurrent testing.

Luckily, the concept of kaizen allows engineers to learn from these past mistakes and delve into a new and promising future, with outlooks into advancing the semiconductor industry, consequently solving aging concerns. Specifically, there is much research being conducted in the field of semiconductor materials and the creation of more effective options. Carbon nanotube Field-Effect Transistors (CNFET) are a candidate to hit the industry, as the "shrinking [of] silicon-based transistors [is] reaching an end" [12]. As they are made of carbon, they theoretically have higher carrier mobility and lower power consumption than their silicon counterparts, making them more energy efficient. Additionally, the CNFET is much more scalable, allowing it to be much smaller than a typical transistor and therefore making it more power dense. Despite ongoing research efforts, uncertainties remain regarding the effective doping of tiny carbon nanotubes with metals to generate positive and negative substrates, even though they are somewhat compatible with existing MOS processes.This makes large-scale production of these transistors a challenge as the CNFET's production is currently nowhere near the time or cost-effectiveness of current technologies. Although this is the case, CNFET is still a very promising technology and illustrates what can be expected in the future regarding material improvements. Beyond materials, transistors themselves are also being highly advanced upon. Specifically, nano-sheet FETs (NSFETs) are becoming the new commercial norm for chipmakers

to create as the scalability of the FinFET, the current FET used by most major electronic device developers, comes to an end. Essentially, an NSFET "is a FinFET on its side with a gate wrapped around it, and it contains several separated horizontal sheets that are vertically stacked, with each sheet containing a channel. A gate surrounds each sheet to form a gate-all-around transistor" [12]. A comparison between a FinFET and a gate-all-around such as a NSFET can be seen in Figure 8, illustrating that while a FinFET has only 3 gates on the channel, a NSFET has 4. This allows for more current control and therefore less current leakage, as it is controlled on four sides. This makes NSFETs not only more scalable, but also less prone to damage, and overall significantly more power and energy efficient. Similar to that of CNFETs, NSFETs are also mostly still in their research phase, as they still face several hurdles such as poor gate length control and being prone to fabrication process variations, aspects which are necessary for the reliability of a semiconductor device. Although this is the case, they are in fact starting to hit the industry to some extent, with "Samsung's multi bridge channel FET (MBCFET) [...] mark[ing] the beginning of NSFETs in the industrial world" [12]. While the aforementioned are just a few of thousands of new semiconductor concepts with potential to reach the market, these products only emphasize the bright future that is in store for technology and semiconductors.

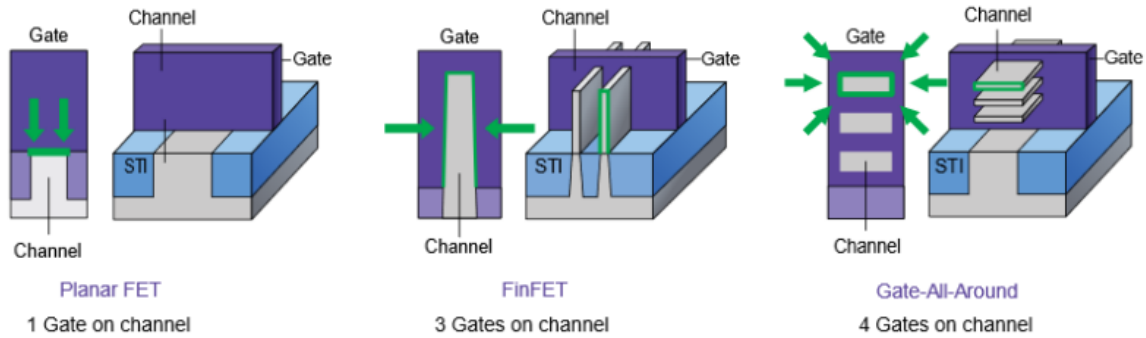

Figure 8: Typical planer FET (left) compared to a FinFET (middle) compared to a gate-all-around (right). [12]

Conclusions:

For the foreseeable future, semiconductors will only become more integral in everyday society as humans continue to advance technologically, and therefore the pressures for improvement will only become greater. The aging of these devices is inevitable; everything ages. If humans knew how to completely prevent aging processes, then they themselves wouldn't die of sickness and old age. So, although this issue may be one that is never "completely" solved, it is theoretically possible to reach device-aging rates that are so minimal that they become negligible relative to the human lifespan. As engineers, it is only nature to desire efficiency in all aspects of life, and in today's world there may be no better way to do this than tackling the issue of semiconductor device aging. While many different forms of aging take place, it is also recognized just as many more mitigation techniques and approaches are practiced in order to address the mounting number of concerns. Not only this, but it is clear that measurement and testing processes for these devices are only becoming more effective over time, and with more

accurate testing comes a more reliable, higher performing product. The future for semiconductors is very bright, and therefore it is paramount that the next generation of young engineers do not let the industry down and become prepared to take on the reins of continuously maximizing the efficiency of semiconductor devices. All in all, understanding and taking action upon semiconductor device aging is critical for maintaining the reliability, performance and integrity of electronic devices, spurring ongoing innovation, and meeting the continuously evolving demands of our modern technology-driven world society.

References:

[1] Kraak, Daniël, et al. "Device aging: A reliability and security concern." *2018 IEEE 23rd European Test Symposium (ETS)*. IEEE, 2018

[2] Zhang, Jian Fu, et al. "Bias temperature instability of mosfets: Physical processes, models, and prediction." *Electronics* 11.9 (2022): 1420.

[3] Acovic, Alexander, Giuseppe La Rosa, and Yuan-Chen Sun. "A review of hot-carrier degradation mechanisms in MOSFETs." *Microelectronics Reliability* 36.7-8 (1996): 845-869.

[4] Wu, Zhicheng, et al. "Investigation of the impact of hot-carrier-induced interface state generation on carrier mobility in nMOSFET." *IEEE Transactions on Electron Devices* 68.7 (2021): 3246-3253.

[5] Liu, Tianshi, et al. "Time-dependent dielectric breakdown of commercial 1.2 kV 4H-SiC power MOSFETs." *IEEE Journal of the Electron Devices Society* 9 (2021): 633-639.

[6] d'Heurle, Francois M. "Electromigration and failure in electronics: An introduction." *Proceedings of the IEEE* 59.10 (1971): 1409-1418.

[7] Sands, Austin, and Vince Tseng. "Game console failure rates." *Project Management Journal* 39.4 (2009).

[8] Austen-Smith, David, et al. "Unintended acceleration: Toyota's recall crisis." *Kellogg School of Management Cases* (2017): 1-16.

[9] Celaya, José R., et al. "Accelerated aging system for prognostics of power semiconductor devices." *2010 Ieee Autotestcon*. IEEE, 2010.

[10] Khoshavi, Navid, et al. "Contemporary CMOS aging mitigation techniques: Survey, taxonomy, and methods." *Integration* 59 (2017): 10-22.

[11] Stratakos, Ioannis, et al. "Aging Evaluation and Mitigation Techniques Targeting FPGA Devices." *Low Power Circuits for Emerging Applications in Communications, Computing, and Sensing*. CRC Press, 2018. 131-149.

[12] Kamal, Kamal Y. "The Silicon Age: Trends in Semiconductor Devices Industry." *Journal of Engineering Science & Technology Review* 15.1 (2022).